Nolan wins funds to 'map' lineages in ovarian cancer cells

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[redacted] PhD, professor of microbiology and immunology, is the first recipient of the Ovarian Cancer Research Program's Teal Innovator Award. The $3.2 million, five-year award, which is administered by the Department of Defense, is intended to advance the understanding and treatment of ovarian cancer.

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The OCRP is one of several Congressionally Directed Medical Research Programs that have arisen since the early 1990s. The programs represent a partnership among the DOD, Congress and the public to fund research into specific diseases or medical conditions. More than 90 research programs have been funded so far, focusing on topics as diverse as Gulf War illness, multiple sclerosis, breast cancer, ovarian cancer, spinal cord injuries and many others.

[redacted] work focuses on the use of an innovative variation on a common cell-sorting technique called flow cytometry. He has devised a way — which he terms single-cell mass cytometry — to measure dozens of biological parameters, including cell size, DNA content and viability, in individual human cells. He plans to use mass cytometry to identify family trees and lineage relationships among tumor cells in individual patients. The knowledge may one day be used to personalize ovarian cancer treatments.

"My laboratory and those of our collaborators are firmly committed to a large-scale effort in this malignancy," Nolan said in an announcement by the OCRP. "Basically, our message is that cancers can be organized, can be mapped, and we can finally understand which cells a given drug has activity against and map this to the molecular biology of the disease. We hope that our work can overcome one or both of the major barriers (deeper classification and early detection) to improving therapeutic outcome for ovarian cancer patients."

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